

IMPACT OF MACROECONOMIC VARIABLES ON STOCK MARKET INDEX ON POLAND

Ani Guraspashvili,

Elza Nodzadze,

Ilia Tschanturia

ABSTRACT

Purpose – This paper's goal is to determine how the stock market and macroeconomic factors are related.

Design/methodology/approach – In order to investigate the connections between inflation, the Industrial Production Index, the exchange rate, the volatility index, and the stock return index, this study used four hypotheses. Secondary data is used. Monthly observations of the variables were made between January 2017 and December 2022. The information was taken from the database of the International Monetary Fund.

Results: The results highlight how important macroeconomic factors are in determining the volatility and behavior of the stock market. Important insights into the relationships between the stock market price, industrial output index, inflation, federal funds rate, VIX index, and exchange rates have been gained by examining earlier research. **Paper type** Research paper

Keywords – Stockreturn index, Stock market, inflation, Industrial Production Index, Exchange rate, Volatility index,

აბსტრაქტი

მიზანი – ამ ნაშრომის მიზანია მაკროეკონომიკურ ცვლადებსა და საფონდო ბაზარს შორის ურთიერთობის დადგენა.

დოზინი/მეთოდოლოგია/მიდგომა – კვლევამ მიიღო ოთხი ჰიპოთეზა და გამოიკვლია ინფლაციას, სამრეწველო წარმოების ინდექსს, გაცვლით კურსს, მერყეობის ინდექსსა და საფონდო შემოსავლის ინდექსს შორის ურთიერთობები. გამოყენებულია მეორადი მონაცემები. ცვლადები დაკვირვებულია ყოველთვიურად 2017 წლის იანვრიდან 2022 წლის დეკემბრამდე. მონაცემები მიღებულია საერთაშორისო სავალუტო ფონდის მონაცემთა ბაზიდან.

შედეგები – კვლევის შედეგები ხაზს უსვამს მაკროეკონომიკური მაჩვენებლების მნიშვნელობას საფონდო ბაზრის ქცევასა და მერყეობაზე გავლენის მხრივ. წინა კვლევების ანალიზით მიღებულია მნიშვნელოვანი ცოდნა საფონდო ბაზრის ფასებს, სამრეწველო წარმოების ინდექსს, ინფლაციას, ფედერალური ფონდების განაკვეთს, VIX ინდექსსა და გაცვლით კურსებს შორის კავშირებზე.

ნაშრომის ტიპი – სამეცნიერო კვლევითი ნაშრომი

INTRODUCTION

Cointegration of stock markets has been the focus of extensive research, especially in industrialized nations. However, research on emerging markets is also becoming more and more popular among academics, particularly in the wake of the 1997 Asian financial crisis (Geetha, et al., 2011). Several studies, like those by Moon (2001) and Elyasiani (1998), have looked at the integration of stock markets by mixing both developed and emerging nations (Moon, 2001) (Perera & Puri, 1998).

The reviewed literature (Ibrahim & Aziz, 2003) (Jefferis & Okeahalam, 2000) (Liu & Shrestha, n.d.) encompasses several key macroeconomic variables and their connection to stock market returns. These variables include stock market price, industrial production index (IPI), inflation (CPI), federal funds rate (FFR), VIX index, and exchange rates. By analyzing the findings of previous studies, a deeper understanding of the implications of these macroeconomic factors on stock market behavior and volatility can be gained (Ibrahim, 1999).

Summarizing the key conclusions of earlier studies and examining the connections between stock market returns and macroeconomic factors are the objectives of this literature review. Important insights into the dynamics and interdependencies between stock markets and macroeconomic indicators can be gained by examining the methodology employed and the empirical data provided. This thorough analysis will add to the amount of knowledge already available in this sector and serve as a basis for future research.

1. LITERATURE REVIEW

Researchers and practitioners have been interested in the connection between stock market returns and macroeconomic variables in prior studies. These kinds of studies have demonstrated that there is a strong correlation between stock markets and macroeconomic factors in different nations. In conclusion, these research can be separated into two main groups based on their areas of interest (Edin & Djedović, 2018).

1.1 STOCK MARKET PRICE

Researchers and practitioners have been interested in the interplay between stock market returns and volatility and macroeconomic variables in earlier studies. These kinds of studies have demonstrated that there is a strong correlation between stock markets and macroeconomic factors in different nations (Edin & Djedović, 2018). Understanding which variables influence the prediction of other variables is the essence of forecasting. This calls for a thorough grasp of the timing of lead-lag relationships among a variety of variables, a comprehension of their statistical significance, and an understanding of which variables are more crucial to monitor as signals for forecasting market movements (Guresen, et al., 2011).

1.2 INDUSTRIAL PRODUCTION INDEX (IPI)

Investors believe that the country's macroeconomic events have a significant impact on the volatility of stock prices, which implies that macroeconomic variables can affect investors' investment decisions and motivates many researchers to look at the correlations between stock returns and macroeconomic variables. For academics, practitioners, and policymakers, identifying the relationship between sector index and IPI has become essential (Amarasinghe, 2016). This, the factor in determining the state of the business cycle is the indicator of industrial production movement. It can be used to project GDP and other metrics because, in theory, it moves parallel to GDP (Edin & Djedović, 2018).

1.3 INFLATION (CPI)

The conventional explanation, money illusion, has actually only a very weak explanatory power, according to a detailed re-examination of alternative explanations for the unusually high correlation between the nominal bond yields and "real" equity yields in US post-war data. The very high prevalence of stagflations accounts for a significant portion of this covariation. There is speculation that during recessions, risk aversion and economic uncertainty may both rise, raising equity risk premiums and, consequently, stock yields (Geert & Eric, 2010).

1.4 FEDERAL FUND RETURN (FFR)

Federal funds rates have an impact on stock market activity. It is discovered that all of the macroeconomic variables examined, with the exception of the industrial output index, significantly affect stock market returns (Slah, et al., 2016).

1.5 VOLATILITY INDEX (VIX)

Researchers contend that the VIX index was designed to be a gauge of stock market volatility, in fact it monitors investor mood. There are identified that

unusual stock market returns near VIX peaks and discuss our results in the context of behavioral finance theory. Evidence supports the theoretical predictions that during the event window surrounding a VIX peak, the prices of large and value stocks experience a relatively minor downward correction, whereas the prices of smaller, more speculative small and growth stocks experience a greater downward correction. According to the additional testing, a group of conventional risk variables cannot account for these cross-sectional variances (Valeriy, 2016).

1.6 EXCHANGE RATE

The stock market index could be significant from a macroeconomic perspective due to its wealth implications on consumption and aggregate demand, and the real exchange rate is a crucial variable given the importance of commerce in the economy (Michelis & Ning, 2010). If industry portfolios with weak factor structures are utilized as the testing assets, or if more reliable methods are used to assess firm-specific exchange rate sensitivity, exchange rate risk as measured by contemporaneous exchange rate movements is not priced in the market (Ding & ou, 2012).

2. METHODOLOGY

In this study, growth rates of the macroeconomic were chosen as the independent variables, while the stock return index were chosen as the dependent variables. The data and empirical techniques used to examine the correlation between the chosen variables are described in this section.

In the research there is used secondary data. These two variables have been observed on a monthly basis from January 2017 till December 2022. The International Monetary Fund's database was used to get the information.

This research adopted four hypothesis and explored the relationships among independent variable: CPI, Exchange rate, Industrial Production Index, Volatility index and dependent variable as Stock return index.

The study used unit root tests for the variables. The unit root test is one of the most relevance in time series analysis. To find the relationship between one or more independent variables and a dependent variable there is used the ordinary least squares regression statistical method.

In the research there is utilized residual diagnostic method: normality assumption, serial correlation and heteroscedasticity.

3. ANALYSIS

3.1 THE ADF UNIT ROOT TESTS

The unit root test is of utmost relevance in time

series analysis because the sequence of integration of the series affects the tools and procedures chosen for further analysis and modeling. The study could yield erroneous findings if the unit root in the variables is not taken into account.¹⁵ To determine the order of integration for the Index return, consumer price index (CPI), DCPI, exchange rate, industrial production index (IPI), DIPI, and Cboe Volatility Index (VIX), the Augmented Dickey-Fuller (ADF) test was initially carried out. The test's findings are displayed in Tables 1, 2, 3, 4, and 5, demonstrating that the series is stationary at first difference.

Table 1. ADF Unit root test for Index return.

Index-return has P-value probability 0.0000, it is smaller than threshold 0.05. this means that it is stationary data.

Table 2. ADF Unit root test for CPI.

CPI (consumer price Index) has P-value probability 1.0000, it is greater than threshold 0.05, so this means that it is non-stationary data.

Table 3. ADF Unit root test for DCPI.

DCPI (consumer price Index) has P-value probability 0.0425, it is smaller than threshold 0.05, so this means that it is stationary data.

Table 4. ADF Unit root test for DCPI.

Exchange-Rate has P-value probability 0.090, it is smaller than threshold 0.05, so this means that it is stationary data.

Table 5. ADF Unit root test for IPI.

IPI (industrial production Index) has P-value probability 0.8326, it is greater than threshold 0.05, so this means that it is non-stationary data.

Table 6. ADF Unit root test for DIPI.

DIPI (industrial production Index) has P-value probability 0.0000, it is smaller than threshold 0.05, so this means that it is stationary data.

Table 7. ADF Unit root test for VIX.

VIX (Cboe Volatility Index) has P-value probability 0.0069, it is smaller than threshold 0.05, so this means that it is stationary data.

3.2 OLS REGRESSION ANALYSIS

The ordinary least squares (OLS) regression statistical method minimizes the sum of squares in the difference between the observed and predicted values of the dependent variable plotted as a straight line in order to

Null Hypothesis: INDEX RETURN has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-8.332386	0.0000
Test critical values: 1% level	-3.527045	
5% level	-2.903566	
10% level	-2.589227	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(INDEX RETURN)
Method: Least Squares
Date: 07/06/23 Time: 18:15
Sample (adjusted): 2017M03 2022M12
Included observations: 70 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
INDEX RETURN(-1)	-1.005303	0.120650	-8.332386	0.0000
C	-0.002062	0.008141	-0.253329	0.8008
R-squared	0.505198	Mean dependent var	-0.000428	
Adjusted R-squared	0.497921	S.D. dependent var	0.096095	
S.E. of regression	0.068091	Akaike info criterion	-2.507796	
Sum squared resid	0.315272	Schwarz criterion	-2.443553	
Log likelihood	89.77285	Hannan-Quinn criter.	-2.482278	
F-statistic	69.42866	Durbin-Watson stat	1.992260	
Prob(F-statistic)	0.000000			

Table 1.

Null Hypothesis: CPI has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	6.151252	1.0000
Test critical values: 1% level	-3.525618	
5% level	-2.902953	
10% level	-2.588902	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(CPI)
Method: Least Squares
Date: 07/06/23 Time: 18:16
Sample (adjusted): 2017M02 2022M12
Included observations: 71 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
CPI(-1)	0.047523	0.007726	6.151252	0.0000
C	-5.091007	0.926268	-5.496254	0.0000
R-squared	0.354162	Mean dependent var	0.583876	
Adjusted R-squared	0.344802	S.D. dependent var	0.862272	
S.E. of regression	0.697960	Akaike info criterion	2.146456	
Sum squared resid	33.61326	Schwarz criterion	2.210193	
Log likelihood	-74.19918	Hannan-Quinn criter.	2.171802	
F-statistic	37.83791	Durbin-Watson stat	1.673930	
Prob(F-statistic)	0.000000			

Table 2.

Null Hypothesis: DCPI has a unit root
Exogenous: Constant
Lag Length: 1 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.973282	0.0425
Test critical values: 1% level	-3.528515	
5% level	-2.904198	
10% level	-2.589562	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DCPI)
Method: Least Squares
Date: 07/06/23 Time: 18:19
Sample (adjusted): 2017M04 2022M12
Included observations: 69 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DCPI(-1)	-0.349927	0.117691	-2.973282	0.0041
D(DCPI(-1))	-0.328284	0.116634	-2.814654	0.0064
C	0.214735	0.111930	1.918467	0.0594
R-squared	0.339484	Mean dependent var	0.003827	
Adjusted R-squared	0.319468	S.D. dependent var	0.885757	
S.E. of regression	0.730699	Akaike info criterion	2.252875	
Sum squared resid	35.23881	Schwarz criterion	2.350010	
Log likelihood	-74.72419	Hannan-Quinn criter.	2.291412	
F-statistic	16.96093	Durbin-Watson stat	1.994764	
Prob(F-statistic)	0.000001			

Table 3.

Null Hypothesis: IPI has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.727352	0.8326
Test critical values: 1% level	-3.525618	
5% level	-2.902953	
10% level	-2.588902	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(IPI)
Method: Least Squares
Date: 07/06/23 Time: 18:21
Sample (adjusted): 2017M02 2022M12
Included observations: 71 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
IPI(-1)	-0.024269	0.033366	-0.727352	0.4695
C	4.467651	5.067184	0.881683	0.3810
R-squared	0.007609	Mean dependent var	0.806843	
Adjusted R-squared	-0.006774	S.D. dependent var	4.929927	
S.E. of regression	4.946595	Akaike info criterion	6.063041	
Sum squared resid	1688.348	Schwarz criterion	6.126778	
Log likelihood	-213.2379	Hannan-Quinn criter.	6.088387	
F-statistic	0.529041	Durbin-Watson stat	1.747069	
Prob(F-statistic)	0.469471			

Table 5.

Null Hypothesis: EXCHANGE RATE has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-3.562862	0.0090
Test critical values: 1% level	-3.525618	
5% level	-2.902953	
10% level	-2.588902	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(EXCHANGE RATE)
Method: Least Squares
Date: 07/06/23 Time: 18:21
Sample (adjusted): 2017M02 2022M12
Included observations: 71 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
EXCHANGE RATE(-1)	-0.339217	0.095209	-3.562862	0.0007
C	31.61533	8.838271	3.577095	0.0006
R-squared	0.155385	Mean dependent var	0.129363	
Adjusted R-squared	0.143144	S.D. dependent var	1.211339	
S.E. of regression	1.121294	Akaike info criterion	3.094609	
Sum squared resid	86.75374	Schwarz criterion	3.158346	
Log likelihood	-107.8586	Hannan-Quinn criter.	3.119955	
F-statistic	12.69398	Durbin-Watson stat	1.710226	
Prob(F-statistic)	0.000670			

Table 4.

Null Hypothesis: DIPI has a unit root
Exogenous: Constant
Lag Length: 1 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-6.986568	0.0000
Test critical values: 1% level	-3.528515	
5% level	-2.904198	
10% level	-2.589562	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(DIPI)
Method: Least Squares
Date: 07/06/23 Time: 18:23
Sample (adjusted): 2017M04 2022M12
Included observations: 69 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DIPI(-1)	-1.109766	0.158843	-6.986568	0.0000
D(DIPI(-1))	0.249895	0.119131	2.097647	0.0398
C	0.870825	0.600385	1.450444	0.1517
R-squared	0.478897	Mean dependent var	-0.013924	
Adjusted R-squared	0.463106	S.D. dependent var	6.657118	
S.E. of regression	4.877873	Akaike info criterion	6.049800	
Sum squared resid	1570.381	Schwarz criterion	6.146935	
Log likelihood	-205.7181	Hannan-Quinn criter.	6.088337	
F-statistic	30.32722	Durbin-Watson stat	2.076821	
Prob(F-statistic)	0.000000			

Table 6.

Null Hypothesis: VIX has a unit root
Exogenous: Constant
Lag Length: 0 (Automatic - based on SIC, maxlag=11)

	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-3.654837	0.0069
Test critical values: 1% level	-3.525618	
5% level	-2.902953	
10% level	-2.588902	

*MacKinnon (1996) one-sided p-values.

Augmented Dickey-Fuller Test Equation
Dependent Variable: D(VIX)
Method: Least Squares
Date: 07/06/23 Time: 18:29
Sample (adjusted): 2017M02 2022M12
Included observations: 71 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
VIX(-1)	-0.317834	0.086963	-3.654837	0.0005
C	6.498346	1.884915	3.447554	0.0010
R-squared	0.162193	Mean dependent var	0.136338	
Adjusted R-squared	0.150050	S.D. dependent var	6.608685	
S.E. of regression	6.092726	Akaike info criterion	6.479833	
Sum squared resid	2561.370	Schwarz criterion	6.543570	
Log likelihood	-228.0341	Hannan-Quinn criter.	6.505179	
F-statistic	13.35783	Durbin-Watson stat	2.130697	
Prob(F-statistic)	0.000498			

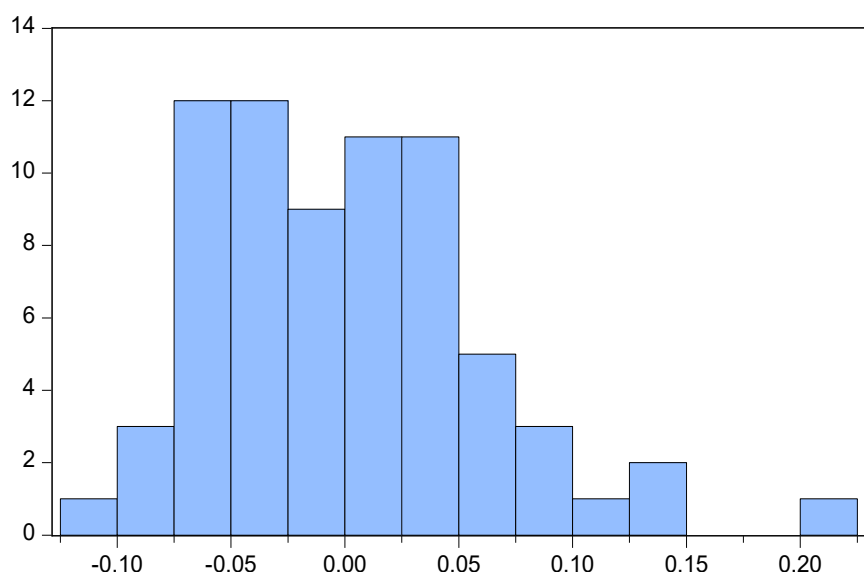
Table 7.

estimate the relationship between one or more independent variables and a dependent variable. OLS regression will be discussed in this article when it comes to bivariate models, which are models where just one independent variable (X) predicts a dependent variable (Y).

Table 8. OLS Regression analysis results
95% CONFIDENCE INTERVAL P-value 0.05

3.3 STATED HYPOTHESIS

H1. There is a statistically significant impact of CPI on index return in Poland.



Series: Residuals
Sample 2017M02 2022M12
Observations 71

Mean 5.46e-17
Median -0.005324
Maximum 0.215751
Minimum -0.104794
Std. Dev. 0.059712
Skewness 0.904831
Kurtosis 4.277319

Jarque-Bera 14.51482
Probability 0.000705

Dependent Variable: INDEX RETURN
Method: Least Squares
Date: 07/06/23 Time: 18:35
Sample (adjusted): 2017M02 2022M12
Included observations: 71 after adjustments

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DCPI	0.003919	0.008852	0.442672	0.6595
DIPI	-0.001365	0.001526	-0.894681	0.3742
EXCHANGE RATE	-0.002061	0.005113	-0.403148	0.6881
VIX	-0.003902	0.000931	-4.189312	0.0001
C	0.267975	0.474142	0.565177	0.5739

R-squared	0.219989	Mean dependent var	-0.001081
Adjusted R-squared	0.172715	S.D. dependent var	0.067610
S.E. of regression	0.061495	Akaike info criterion	-2.671907
Sum squared resid	0.249586	Schwarz criterion	-2.512564
Log likelihood	99.85272	Hannan-Quinn criter.	-2.608542
F-statistic	4.653537	Durbin-Watson stat	1.629906
Prob(F-statistic)	0.002271		

Table 8.

H2. There is a statistically significant impact of IPI on index return in Poland.

H3. There is a statistically significant impact of Exchange rate on index return in Poland.

H4. There is a statistically significant impact of VIX on index return in Poland.

As it is shown in the table H1, H2 and H3 has P-value greater than 0.05, so this means that these three hypotheses are rejected. And the H4. has P-value smaller than 0.05, so this hypothesis is accepted.

3.4 RESIDUAL DIAGNOSTICS

In the research there is histogram of residuals for detecting normality assumption. As the table below shows that probability P-value is 0.000705, this means that data is non-normally distributed. In case, it would have been greater than threshold 0.05, than data would have been normally distributed.

30+ observations

3.5 SERIAL CORRELATION

The relationship that develops over time between a given variable and its lagged form is known as serial correlation. It assesses the difference between the current and past values of a variable. The P-value in the table below is significantly higher than the cutoff of 0.05. Consequently, the variables' current and past values do not exhibit serial correlation.

Breusch-Godfrey Serial Correlation LM Test:

F-statistic	1.371117	Prob. F(2,64)	0.2612
Obs*R-squared	2.917173	Prob. Chi-Square(2)	0.2326

Test Equation:

Dependent Variable: RESID

Method: Least Squares

Date: 07/06/23 Time: 18:39

Sample: 2017M02 2022M12

Included observations: 71

Presample missing value lagged residuals set to zero.

Variable	Coefficient	Std. Error	t-Statistic	Prob.
DCPI	0.003711	0.009105	0.407536	0.6850
DPI	-0.001089	0.001683	-0.646990	0.5200
EXCHANGE RATE	-0.001625	0.005191	-0.312934	0.7553
VIX	-0.000365	0.000962	-0.379543	0.7055
C	0.157347	0.481989	0.326454	0.7451
RESID(-1)	0.227113	0.139375	1.629514	0.1081
RESID(-2)	0.031908	0.135064	0.236240	0.8140

R-squared	0.041087	Mean dependent var	5.46E-17
Adjusted R-squared	-0.048811	S.D. dependent var	0.059712
S.E. of regression	0.061152	Akaike info criterion	-2.657524
Sum squared resid	0.239332	Schwarz criterion	-2.434443
Log likelihood	101.3421	Hannan-Quinn criter.	-2.568812
F-statistic	0.457039	Durbin-Watson stat	1.969438
Prob(F-statistic)	0.837365		

3.6 HETEROSCEDASTICITY

The statistical word for the error variance, or scattering dependency, in at least one independent variable

Heteroskedasticity Test: Breusch-Pagan-Godfrey

F-statistic	0.633129	Prob. F(4,66)	0.6406
Obs*R-squared	2.623699	Prob. Chi-Square(4)	0.6226
Scaled explained SS	3.715129	Prob. Chi-Square(4)	0.4459

Test Equation:

Dependent Variable: RESID^2

Method: Least Squares

Date: 07/06/23 Time: 18:41

Sample: 2017M02 2022M12

Included observations: 71

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	-0.038494	0.049943	-0.770763	0.4436
DCPI	-2.42E-05	0.000932	-0.025917	0.9794
DPI	-9.70E-05	0.000161	-0.603289	0.5484
EXCHANGE RATE	0.000430	0.000539	0.798354	0.4275
VIX	0.000106	9.81E-05	1.080171	0.2840

R-squared	0.036954	Mean dependent var	0.003515
Adjusted R-squared	-0.021413	S.D. dependent var	0.006409
S.E. of regression	0.006477	Akaike info criterion	-7.173165
Sum squared resid	0.002769	Schwarz criterion	-7.013821
Log likelihood	259.6474	Hannan-Quinn criter.	-7.109799
F-statistic	0.633129	Durbin-Watson stat	1.740438
Prob(F-statistic)	0.640631		

within a sample is heteroscedasticity, or heteroscedasticity. Table below shows P value of 0.640431, so that means that there is no heteroscedasticity problem (homoscedastic).

DISCUSSION

The reviewed studies consistently demonstrate a strong correlation between stock market returns and macroeconomic variables. Researchers have employed various methodologies and statistical approaches to investigate this relationship, leading to valuable insights and findings. The results highlight the importance of macroeconomic indicators in predicting and understanding stock market behavior.

The stock market price has been a primary focus of many studies, and its connection to macroeconomic factors has been well-established. Researchers have examined the interplay between stock market returns, volatility, and different macroeconomic variables. The evidence indicates a robust correlation, suggesting that changes in macroeconomic conditions significantly influence stock market movements. Understanding the timing, significance, and relative importance of these variables is essential for effective market forecasting.

The industrial production index (IPI) has emerged as a critical macroeconomic indicator, serving as a barometer for the state of the business cycle. Its parallel movement with GDP makes it a valuable tool for projecting economic growth and identifying turning points in the economy. Moreover, inflation, as measured by the Consumer Price Index (CPI), has also been investigated in relation to stock market returns. While the conventional explanation of money illusion offers limited explanatory power, researchers have found that stagflation and recessions play a significant role in the correlation between nominal bond yields, equity yields, and stock market performance.

The federal funds rate (FFR), as a monetary policy tool, has been found to have a discernible impact on stock market activity. Researchers have observed the influence of various macroeconomic variables on stock market returns, except for the industrial output index. These findings highlight the sensitivity of stock markets to changes in interest rates and underline the importance of monetary policy decisions in shaping market dynamics.

The VIX index, designed as a gauge of stock market volatility, has been a subject of interest in behavioral finance. Researchers have identified that unusual stock market returns occur around VIX peaks, indicating investor sentiment and risk aversion. The findings support theoretical predictions, showing that prices of large and value stocks experience minor downward corrections during VIX peaks, while smaller and more

speculative stocks face greater downward corrections. Importantly, these cross-sectional variances cannot be accounted for by conventional risk variables, underscoring the unique nature of VIX-related market movements.

Lastly, the connection between stock market returns and exchange rates has been explored due to its implications for wealth, consumption, and aggregate demand. The real exchange rate, in particular, holds significance given its relevance to international trade and commerce. However, studies investigating the pricing of exchange rate risk in the stock market have yielded mixed results. Further research and improved measurement approaches are needed to enhance understanding of this relationship.

CONCLUSION

In conclusion, this comprehensive literature review highlights the strong correlation between stock market returns and macroeconomic variables. The findings underscore the significance of macroeconomic indicators in influencing stock market behavior and volatility. By analyzing previous studies, valuable insights into the connections between stock market price, industrial production index (IPI), inflation (CPI), federal funds rate (FFR), VIX index, and exchange rates have been obtained.

These findings have important implications for investors, policymakers, and researchers alike. Understanding the relationships between stock markets and macroeconomic variables can aid in forecasting market movements, making informed investment decisions, and formulating effective monetary and fiscal policies. Future research should continue to explore these connections, employing robust methodologies and refining measurement approaches to deepen our understanding of the complex interplay between stock markets and macroeconomic factors. By doing so, the ability to navigate and respond to the dynamics of the global financial landscape can be enhanced.

BIBLIOGRAPHY

1. Ibrahim, M., 1999. Macroeconomic Variables and Stock returns in Malaysia: An Empirical Analysis. *Asian Economic Journal*, pp. 219-231.
2. Amarasinghe, A., 2016. A study on the impact of industrial production index (IPI) to. *International Conference of Sabaragamuwa University of Sri Lanka 2015*, pp. 275-278.
3. Ding , D. & ou, h., 2012. Exchange rate risk in the US stock market. *Journal of International Financial*, pp. 137-150.
4. Edin , D. & Djedović, E., 2018. IMPACT OF MACRO-ECONOMIC. *GODINA IV • BROJ 2* .
5. Geert , B. & Eric , E., 2010. Inflation and the stock market: Understanding the “Fed Model”\$. *Journal of Monetary Economics*, pp. 278-294.
6. Geetha, C., Mohidin, R., Chandran , V. V. & Chong, V., 2011. THE RELATIONSHIP BETWEEN INFLATION AND STOCK MARKET: EVIDENCE FROM MALAYSIA, UNITED STATES AND CHINA. *International Journal of Economics and Management Sciences* , 1(2), pp. 01-16.
7. Guresen , E., Gulgun , K. & Tugrul U. , . D., 2011. Using artificial neural network models in stock market index prediction. *Expert Systems with Applications*, Volume 38, pp. 10389-10397.
8. Ibrahim, M. & Aziz, H., 2003. Macroeconomic variables and the Malaysian equity market: A view through rolling subsamples. . *J. Econ. Stud*, pp. 6-27.
9. Jefferis, K. & Okeahalam, C., 2000. “The Impact of Economic Fundamentals on Stock Markets in Southern Africa. *Developmentouthern Africa*, pp. 23-51.
10. Liu, M. & Shrestha, K. M., n.d. Analysis of the long-term relationship between macroeconomic variables and the Chinese stock market using heteroskedasticity cointegration. *Managerial Finance*, Volume 744-755, p. 2008.
11. Michelis, L. & Ning, C., 2010. The dependence structure between the Canadian stock market and the USD/CAD Canadian stock market and the USD/CAD. *Canadian Journal of Economics*, Volume 43, p. 1016-1039.
12. Moon, W. s., 2001. Currency Crisis and Stock Market Intergration:A comparison between East Eastern and European Experiences. *Journal of International and Area Studies*, pp. 41-56.
13. Perera, E. & Puri, T. N., 1998. Interdependence and Dynamic Linkages between Stock Market in Elyasiani. *Journal of Multinational Financial Management*, pp. 89-101.
14. Slah, B., Mourad , M. & Nader , N., 2016. The impact of macroeconomic and conventional stock market variables on Islamic. *Borsa istanbul Review*.
15. Valeriy , Z., 2016. Abnormal Stock Market Returns around Peaks in VIX: The Evidence of Investor Overreaction?. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2773134.